

LAITH ABOUHASOUN

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Education

Bachelor of Science in Mechanical Engineering — San Diego State University | Graduation: May 2026

Focus Areas: UAV Systems, Autonomous Systems, VTOL Aircraft Design, Rapid Prototyping, CAD Design, Fabrication, Testing & Validation, Mechanical Integration.

Technical Skills

- **Software & Simulation:** SolidWorks • CAD • MATLAB • ANSYS Fluent • FEA & CFD Setup • G-code
 - **Design & Fabrication:** 3D Printing • CNC Machining • Hot-Wire Foam Cutting • Waterjet Cutting
 - **Mechanical Systems:** UAV Integration • Servo Mechanisms • Mechanical Assembly • Rapid Prototyping
 - **Testing & Validation:** Flight Testing • System Integration • Troubleshooting • Prototype Testing
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Engineering Projects

VTOL Unmanned Aerial System — Design & Manufacturing Lead, Safety Pilot

Joint ME/ECE/AE Capstone Project | Sep 2025 – May 2026

- Designed mechanical and structural components for a biplane VTOL UAS, including motor mounts, connectors, and landing structures using SolidWorks.
- Manufactured prototype components using PA6-CF 3D printing, hot-wire foam cutting, and fiberglass layups.
- Performed fit checks, mechanical assembly, and integration of wings, spars, and electronics mounts.
- Conducted thrust-stand testing and collected propulsion performance data to support motor selection and propulsion evaluation.
- Collaborated with aerospace and electrical engineering teammates on system integration, CAD layout, and design reviews, with a focus on mechanical subsystems.

Autonomous UAV Payload Recovery Mechanism — Independent Project

Lead Designer & Manufacturer | Jan 2026 – May 2026

- Independently designed and manufactured an adaptive UAV payload recovery mechanism for autonomous aerial payload capture.
 - Developed and tested a lightweight gripping system capable of handling payloads with varying shapes and sizes.
 - Iterated multiple prototypes to improve strength, reliability, and manufacturability.
 - Enhanced gripping performance and payload retention through innovative mechanical design features.
 - Successfully demonstrated payload capture and retention with loads up to 2.5kg, exceeding mission requirements.
 - Received strong positive feedback from faculty and industry professionals regarding the design's novelty and potential commercial applications.
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Professional Experience

Ali & Fahad Transportation – Fleet Maintenance Technician | January 2024 – Present

- Perform routine mechanical inspections, diagnostics, and preventative maintenance on fleet vehicles.
 - Assist with troubleshooting mechanical, electrical, and hydraulic systems following established maintenance procedures.
 - Replace mechanical components, apply proper torque specifications, and use precision measurement tools during repairs.
 - Maintain service documentation, track parts usage, and support fleet maintenance scheduling.
 - Develop hands-on experience with mechanical systems, fasteners, tools, and safety practices in a professional shop environment.
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Languages

English (Fluent), Arabic (Fluent)